UNIT III
5^{th} stage in Design Thinking - TEST Phase

WHAT is the Test mode?

The Test mode is when you solicit feedback, about the prototypes you have created, from your users and have another opportunity to gain empathy for the people you are designing for. Testing is another opportunity to understand your user, but unlike your initial empathy mode, you have now likely done more framing of the problem and created prototypes to test. Both these things tend to focus the interaction with users, but don't reduce your "testing" work to asking whether or not people like your solution. Instead, continue to ask "Why?", and focus on what you can learn about the person and the problem as well as your potential solutions.

Ideally you can test within a real context of the user's life. For a physical object, ask people to take it with them and use it within their normal routines. For an experience, try to create a scenario in a location that would capture the real situation. If testing a prototype in situ
is not possible, frame a more realistic situation by having users take on a role or task when approaching your prototype. A rule of thumb: always prototype as if you know you're right, but test as if you know you're wrong—testing is the chance to refine your solutions and make them better.

WHY test?

1. **To refine prototypes and solutions**. Testing informs the next iterations of prototypes. Sometimes this means going back to the drawing board.
2. **To learn more about your user.** Testing is another opportunity to build empathy through observation and engagement—it often yields unexpected insights.
3. **To refine your POV**. Sometimes testing reveals that not only did you not get the solution right, but also that you failed to frame the problem correctly.

HOW to test?

1. **Show don't tell**. Put your prototype in the user's hands – or your user within an experience. And don't explain everything (yet). Let your tester interpret the prototype. Watch how they use (and misuse!) what you have given them, and how they handle and interact with it; then listen to what they say about it, and the questions they have.

2. **Create Experiences**. Create your prototypes and test them in a way that feels like an experience that your user is reacting to, rather than an explanation that your user is evaluating.

3. **Ask users to compare**. Bringing multiple prototypes to the field to test gives users a basis for comparison, and comparisons often reveal latent needs.

“Iteration and making the process your own”

Iteration is a fundamental of good design. Iterate both by cycling through the process multiple times, and also by iterating within a step—for example by creating multiple prototypes or trying variations of a brainstorming topics with multiple groups. Generally as you take multiple cycles through the design process your scope narrows and you move from working on the broad concept to the nuanced details, but the process still supports this development.

For simplicity, the process is articulated here as a linear progression, but design challenges can be taken on by using the design modes in various orders;
Furthermore there are an unlimited number of design frameworks with which to work. The process presented here is one suggestion of a framework; ultimately you will make the process your own and adapt it to your style and your work. Hone your own process that works for you. Most importantly, as you continue to practice innovation you take on a designedly mind-set that permeates the way you work, regardless of what process you use.

KANO MODEL

What is Kano Model?

Definition: The Kano model is a customer satisfaction measurement tool that prioritizes features based on customer reaction to their presence or absence, effectively determining their impact on customer satisfaction.

How Does the Kano Model Work?

The model assigns three types of attribute (or property) to products and services:
1. Threshold Attributes (Basics). These are the basic features that customers expect a product or service to have.
For example, when you book into a hotel, you'd expect hot water and a bed with clean linen as an absolute minimum.
2. Performance Attributes (Satisfiers). These elements are not absolutely necessary, but they increase a customer's enjoyment of the product or service.
Returning to our example, you'd be pleased to discover that your hotel room had free superfast broadband and an HD TV, when you'd normally expect to find paid-for wi-fi and a standard TV.
3. Excitement Attributes (Delighters). These are the surprise elements that can really boost your product's competitive edge. They are the features that customers don't even know they want, but are delighted with when they find them.
In your hotel room, that might be finding the complimentary Belgian chocolates that the evening turn-down service has left on the bed.

Figure 1, below, illustrates how the presence (or absence) of each of the three attributes in a product or service can affect customer satisfaction.

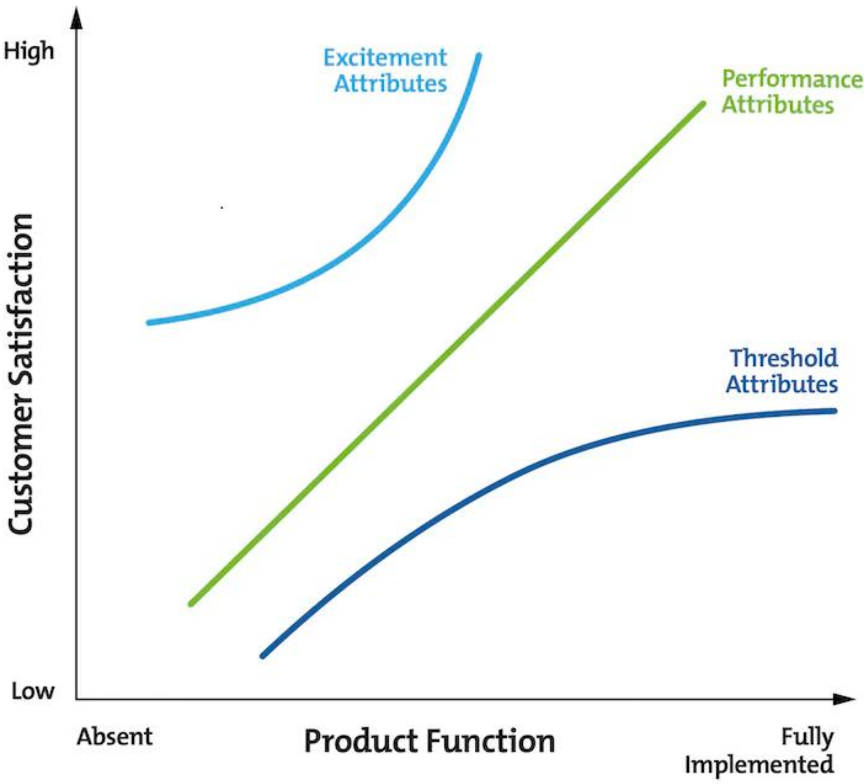

Figure 1 – The Kano Model
Kano Model Analysis - Delivering Products That Will Delight

How to Use the Kano Model

Before you apply Kano Model Analysis, be sure to find out what your customers really value. Never assume that you know! Ask them what they like, what they love, and what they dislike.

Steps to be followed:

1. Research and brainstorm all of the possible features and attributes of your product or service, and everything you can do to please your customers.
2. Classify these as Threshold, Performance or Excitement Attributes and add a fourth type, Not Relevant. These are the things that don't add value because customers don't care about them.
3. Make sure that your product or service has all of the essential Threshold Attributes. If necessary, eliminate some Performance Attributes so that you can include these features.
4. Assess the Excitement Attributes, and think about how you can incorporate some of them into your product or service. Again, if necessary, cut some Performance Attributes, so that you can afford to invest in your Excitement Attribute.
5. Choose the Performance Attributes that you can deliver at a competitive price, while still maintaining an acceptable profit margin.

FUTURE: Design Thinking meets the corporation

3 Key Design Thinking principles that redefine business:

1. Humankind has survived thus far because design can work well together, communicate, empathize, anticipate, understand, and exchange. Design thinking is a reflection of these abilities.
2. The culture behind its practices, principles, and process is potentially more empathetic, human-centered, and courageous than business management.
3. A multifunctional and multi perspective approach to solving problems has influenced many of the principles inherent in design thinking.

The Ten Design Thinking principles that redefine business or business management are:

1. Action -Oriented: - It proposes a cross-disciplinary learning-by-doing approach to problem solving.
- It allows designers to accommodate varied interests and abilities through hands-on and applied learning experiences between individuals.
- A big part of design thinking is design doing.

2. Comfortable with change:
- It is disruptive and provocative by nature because it promotes new ways of looking at problems.
- A large part of the design thinking process is stepping out of conventional roles and escaping from existing dogmas to explore new approaches to problem solving.

3. Human-centric:
- It is always focused on the customer or end user's needs, including unarticulated, unmet, and unknown needs.
- Design Thinking employs various observational and listening-based research techniques to systematically learn about the needs, tasks, steps, and milestones of person's process.

4. Integrates foresight:
- Foresight opens up the future and invites designers to explore uncertainties.
- It encourages designers to be comfortable with working with unknowns and expects designers to cope with inadequate information in the process of discovering and creating a tangible outcome.

5. A Dynamic Constructive Process:
- It is iterative
- It requires ongoing definition, redefinition, representation, assessment, and visualization.
- It is a continuous learning experience arising out of a need to obtain and apply insights to shifting goals.
- Prototyping, creating of tangible sharable artifacts, become an important piece of the design thinking tool.

6. Promotes Empathy:
 design Thinking encourages the use of tools to help designers communicate with people in order to better understand their behaviours, exceptions, values, motivations and the needs that drive them and will improve their lives.
 designers use these insights to develop new knowledge through creative learning and experimentation.

7. Reduces Risk:
 Whether it is developing and launching a new product or service, there are many benefits in learning from small and smart failures.

8. Create Meaning:
 Creating meaning is the hardest part of the design process, and the communication tools used in design thinking-maps, models, sketches and stories -help capture and express the information required to form and socialize meaning.
 Arriving this takes time and emerges through multiple iterations and conversations.

9. Bring Enterprise creativity to next level:
 Design thinking fosters a culture that embraces questioning, inspire frequent reflection in action, celebrates creativity, embraces ambiguity, and creates visual sense making through interactions with visualizations, physical objects and people.
 Design thinking organization creates strong 'inspirationalization" and sensibility to give tangibility to theemotional contract that employees have with organizations.

10.The New Competitive Logic of Business Strategy:
 Design thinking is the most complementary practice that can be applied side by side with Michael porter's theory of competitive strategy.
 It allows companies to create new products, experiences, processes and business models beyond simply what works.
 It turns designers into desirable products, which is a truly sustainable competitive advantage through innovation.

Various business challenges are:

- Growth
- Predictability
- Change
- Maintaining Relevance
- Extreme competition
- Standardization.
- Creative culture
- Strategy and organization

Design thinking to meet corporate needs:

- Design thinking has become a pet phrase for many successful businesses today but its impacts are very circumstantial and differ for each industry
- It helps brands stay ahead of the curve by driving innovation in a business environment.
- A human-centric approach towards problem-solving makes it an effective bridge between brands and customers.
- Experts use it for enhancing both physical and digital experiences of products and services.
- Companies resorting to design thinking consider design much more than a phase or a department – in fact, it shapes the entire thought behind business goals.
- Building a design-optimised company culture will certainly drive more innovation and customer satisfaction.
- If designers are wondering how different industries benefit from design thinking, have compiled a list of case studies to help designers to understand how it can be applied in each context.

Design Activism:

Design activism is about using your talents as a designer to create a positive impact in the world. It means using your talents to help non-profits establish their identity or offering to create a logo for an organization run entirely by volunteers. It's about designers using what they know best, design, to help solve problems and build a better world.

Question Bank:
(Short Answers)

Q.No.	Question	Reference page number (Above)for answer
1	**What is test mode?**	1
2	**List 3 reasons for "why to test"?**	1
3	**List 3 reasons for "how to test"?**	1
4	**Draw the figure of Kano Model?**	3
5	**Give any one key Design Thinking principle that redefines business?**	4
6	**List any 3 Design Thinking principles that redefine business or business management?**	4,5
7	**List any 4 business challenges?**	5
8	**What is design activism?**	6

Question Bank:
(Descriptive Answers)

Q.No.	Question	Reference page number (Above)for answer
1	Explain “Iteration and making the process your own” with respect to test phase of Design thinking?	2
2	Identify the 3 attributes & explain how does the Kano Model Works?	2
3	Explain how to Use the Kano Model?	3
4	Identify and explain any 4 Design Thinking principles that redefine business or business management?	4,5
5	Explain how Design thinking can be applied to meet corporate needs?	6
6	Identify and explain 3reasons for “how to test”	1
7	Apply first 2 stages of Design Thinking to your respective team problem statement and explain both the steps in detail?	As per respective team activity
8	Apply last stages of Design Thinking to your respective team problem statement and explain both the steps in detail?	